SAMPLE

Optics

The Study of Light and Lenses

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Introduction

Optics is the science of light: how it travels, bends, spreads, interferes, and finally registers in eyes and instruments. It is the oldest branch of physics with a continuous written tradition, older than mechanics, older than astronomy's instruments, argued in Greek, refined in Arabic, argued again in Latin and Italian and Dutch, and settled, in stages, in the nineteenth and twentieth centuries. The modern field divides itself by how much of light's nature a given problem needs: rays drawn as straight lines for lenses and mirrors, waves for interference and diffraction, electric and magnetic fields for polarization and scattering, and photons for detection and the laser. Each model is a honest simplification of the one below it, and part of this book's pleasure is watching the tower get built, one era at a time, by people whose names the reader already half knows.

The word light in this book means more than the eye reports. The visible band, the narrow octave of wavelengths the retina answers to, sits inside an enormous spectrum, radio at the long end, infrared just past the red, ultraviolet past the violet, and the X-rays far beyond, and the physics this book describes, the bending, the fringes, the polarization, the photons, belongs to all of it; a radio telescope and a gamma-ray satellite are optical instruments exactly as a hand lens is, working bands where glass turns opaque and mirrors must be tilted sheets of metal or shaded crystals of germanium. The discoveries of 1800 and 1801, the invisible rays on either side of the visible, turned the science of seeing into the science of the whole spectrum, and the modern usage, where optics means the behavior of electromagnetic radiation across all its bands, is the field's own widening of its founding question, from what do we see to what is there to be seen.

The practical inheritance is just as large. Spectacles, the microscope, the telescope, the camera, the laser, and the optical fiber that carries most of the world's information are all optics, and the field's history and its technology grew together, each pushing the other.

A note on the arrangement of the chapters. The first three carry the history to the moment theory and trade met; the middle chapters build the working physics in the field's own ladder, rays, waves, fields, and photons, each chapter assuming only what the earlier ones built; the later chapters apply the physics, to the eye, its illusions, the instruments, and the modern light sources; and the closing chapters weigh the trivia and summarize. The reader who wants one thread through it all can follow a single question, what is light, as each successive chapter sharpens the answer, and the reader who wants the practical shortest path can read the instrument chapters alone; the book is built to serve both, and the chapters cross-reference honestly rather than pretending to be independent essays. This book tells both stories in plain language, with no equations: the ancient lenses and the medieval Arabic revolution in the theory of vision; the Dutch and Italian lens grinders; the great war between waves and particles that Newton lost and Young and Fresnel won; Maxwell's unification of light with electricity; the quantum account of photons and the laser's invention; and the modern landscape from antireflective coatings to optical tweezers that move single cells with nothing but light. The account keeps to verified fact, names its people honestly, and closes each chapter by connecting the physics to things the reader can see, wear, look through, or use.

Chapter 1: The Oldest Lenses and the Riddle of Seeing

The story opens with polished rock. The Nimrud lens, a small oval of ground rock crystal unearthed from an Assyrian palace and dated around seven hundred years before the common era, may have been an ornament, a magnifier, or a burning glass for kindling fire, but its shape is undeniably optical, ground with a care that implies a purpose; lenses of comparable age are associated with the Greek island of Rhodes, a glassworking center whose craftsmen filled glass spheres with water and learned, surely by accident first, that the sphere magnified whatever it touched. The ancient world had no theory of what these objects did, but it had the objects, and it had a second, larger question to argue about: what seeing is.

The Greek debate is the first optics controversy on record, and it divided into two camps. The emission theory, articulated by Plato, held that the eye sends out rays, a kind of invisible fire that reaches out to touch objects, and that vision happens at the meeting; the intromission theory held instead that something comes from the objects into the eye, images or copies flowing inward. Both camps had followers for centuries, and the honest reader should notice that everyday language still carries the emission theory inside it: people say their eyes sweep a room, or that a glance strikes someone, phrases that quietly grant the eye an outbound power it does not have.

The mathematicians built on the emission side, because rays were something they could calculate with. Euclid wrote an optics treatise that treated vision as straight lines from the eye and worked out the geometry of perspective, the apparent size of objects scaling with the inverse of their distance, an approach that made optics a branch of geometry and gave the field its first true theorems. Ptolemy, centuries later, wrote a treatise that held an ingenious compromise, an extramission-intromission theory in which rays leave the eye and meet a flow coming in from the object; he summarized much of Euclid, and he went further, describing an apparatus for measuring the angle of refraction, the bending of light as it enters water, and tabulating real numbers, though he never arrived at the correct law hiding in his own data.

The chapter closes with the observer who joined the two threads.

One practical note belongs beside the debate's history: none of it stopped the useful arts. Whatever seeing was, water spheres magnified, bronze mirrors reflected, and burning glasses kindled, and the artisans' optics ran on results the philosophers had not settled, a standing pattern in the field that recurs in every chapter: the trade runs ahead of the theory, the theory catches up and improves the trade, and the improved trade poses the next theory's questions. The reader should hold the pattern loosely in mind through the whole book, because it is the field's real chronology, and the arguments over what light is, however celebrated, were the pauses between its working centuries. Around the ninth century, al-Kindi, writing in Baghdad, compared the Aristotelian and Euclidean accounts of vision and argued, against the day's fashions, that the object itself must be the source of what is seen; his work prepared the ground for the great resolution that follows in the next chapter, when the riddle of seeing, argued for a thousand years, was finally solved by an experimentalist who simply tested it.

The riddle deserves one last honest note before its solution, because the ancient answers were not foolish. The emission theory explained real experiences, the sense of directed attention, the hurt of the sun, the eye's effort in the dark, and the intromission theory had its own problem, why objects are seen whole and sharp if every point streams copies toward every eye, a difficulty that made eminent people hedge for centuries. The history's lesson is the general one: a wrong theory that explains something persists until the better account explains it and the rest, and the seeing riddle held physics for a millennium precisely because it is not a physics question alone, but a question about mind and world that the physics, when it finally arrived, answered only on its side of the border, the border Alhazen himself drew and the illusion chapter patrols.

The economics of the ancient world also carried optics in ways the texts underplay. Polished metal mirrors, bronze and silver, were luxury goods across the Mediterranean, and the mirror-trade's quality problem, the blurred reflection of a hammer-finned surface, drove the invention of glass mirrors centuries later; burning lenses and mirrors, which concentrate sunlight to a point hot enough to kindle, appear in Greek and Chinese sources both, and the burning glass remained for millennia the most dramatic demonstration that light carries energy and travels in straight lines. Water-filled spheres served as magnifiers for fine engraving and as burning glasses, and at least one classical tradition credits archimedes with mirrors that set ships aflame, a story the folklore column of this book will grade honestly: the physics of a mirror-array igniting a distant hull is marginal, the historical evidence thinner, and yet the legend itself is evidence that the ancients knew concentrated light was concentrated fire.

The debate over seeing also carried a moral the modern reader should keep. The emission theory granted the observer an active, almost sovereign role, the eye reaching out to grasp the world, and the intromission theory demoted the eye to a receiver, a window things write upon. The winning theory's consequences ran past optics: vision became evidence, the incoming light a faithful courier of the object's shape and color, and the whole later edifice of observation-based science quietly stands on Alhazen's side of the argument. When a modern instrument registers a star's spectrum or a cell's image, it is the intromission theory with brass and glass attached, and the ancient argument is over every time an observation is trusted.

A word on the transmission of it all, because the book's geography deserves its map. The Greek texts and Alhazen's Arabic reached Latin Europe through translation movements, first in Spain and then in Italy, and the field's lineage, from Euclid's Alexandria to al-Kindi's Baghdad to Alhazen's Cairo to Grosseteste's Oxford and Bacon's Paris, is the clearest single demonstration in the history of science that knowledge is a relay across languages and centuries, not a national inheritance. The translators get too little credit in the popular tellings, and the reader should know their work was as exacting as the authors': the Latin De aspectibus carried Alhazen across, and its vocabulary, species, radiance, and the geometry of sight, became the scholastics' own, the field's longest quotation.